# PCRQ 10

Graded

**Student**

VK Nguyen

**Total Points**

**4 / 4 pts**

**Question 1**

## True or False on suprema

**1** / 1 pt

✔ **+ 1 pt** Correct

**+ 0 pts** Incorrect

**Question 2**

## (no title)

**1** / 1 pt

✔ **+ 1 pt** Correct

**+ 0 pts** Incorrect

**Question 3**

## (no title)

**1** / 1 pt

✔ **+ 1 pt** Correct

**+ 0 pts** incorrect

**+ 0 pts** blank

**Question 4**

## (no title)

**1** / 1 pt

✔ **+ 1 pt** good-faith answer

**+ 0 pts** Blank

Adding $i$" sounds like adding one element, but a field has to be closed under addition and multiplication. As soon as $i$ is in, so is everything you can build from it with real numbers:

$3i$ (a real times $i$)
$2 + 3i$ (a real plus that)
$(2 + 3i)(1 - i)$, and so on

So "$\mathbb{R}$ plus $i$" really means all of these. This is Remark 2 in the notes: any field containing $\mathbb{R}$ and $i$ must contain every $x + yi$.

Why define the operations explicitly? Writing a new symbol $i$ doesn't tell you what $2 + 3i$ means or how to multiply two such things. You can't just declare "$i^2 = -1$" and hope arithmetic works. That's how mathematicians treated $i$ for centuries, and it's why it seemed mysterious (section 3). The ordered-pair definition makes it rigorous: we define exactly how to add and multiply, then check that all field axioms hold.

The nice surprise: the form $x + yi$ is enough. Adding or multiplying two such expressions gives another one of the same form, since $i^2 = -1$ turns any $i^2$ term back into a real number. So we never need $i^2$, $i^3$, or anything new; adding $i$ and closing up stops right at $x + yi$.

**Q1 True or False on suprema**
**1 Point**

Let $a = \sup(A)$ and let $\varepsilon > 0$. There must exist $x \in A$ such that $a - \varepsilon < x < a$.

- ○ True
- ◉ False

**Q2**
**1 Point**

Recall that multiplication on $\mathbb{C} = \mathbb{R}^2$ is defined by

$$(a, b)(c, d) = (ac - bd, ad + bc).$$

What is

$$(1, 2)(3, -1)?$$

- ○ $(3, -2)$
- ○ $(1, 5)$
- ◉ $(5, 5)$
- ○ $(5, -5)$

**Q3**
**1 Point**

Explain why there is no order on $\mathbb{C}$ that is compatible with the field operations and makes $\mathbb{C}$ an ordered field.

i=√-1 => i^2=-1, but x^2>0 in Lecture 7, so there is a contradiction.

**Q4**

**1 Point**

Please write briefly (in two or three sentences) what single point of the notes of Lecture 11 you found to be the most difficult or confusing. If you did not find any part of it difficult or confusing, please say what part(s) you found most interesting.

I'm confused by the idea of constructing complex numbers from real numbers by adding i, where i^2=-1. If we only need to add i to R, then why we need to define operations between expressions of the form x+yi.